FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Main)-2019

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. Graph is for Isobaric process where pressure and molar heat capacity remains constant V-T graph will be straight line but temperature cannot be zero Kelvin.

2. C

Sol. $f=\frac{nv}{2l}\left(n=2\ for\ first\,overtone\right)$

So, $f=\frac{v}{l}=\frac{\sqrt{T/\mu}}{l}=100\text{Hz}$

3. C

Sol. $I=\frac{4ml^2}{3}+2\left[\frac{ml^2}{12}+\frac{3ml^2}{4}\right]=3ml^2$

4. B

Sol. $w_1t-w_2t=n\left(2\pi\right)$

For minimum t, $n=1$

$t=\frac{T_1T_2}{T_2-T_1}=42\,s$

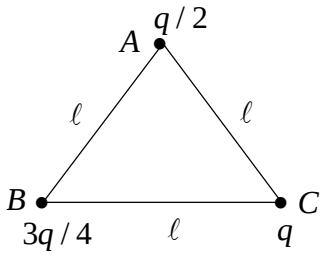
5. D

Sol. Time of flight $=\frac{2u\sin\theta}{g}$ *and* $g=\frac{GM}{R^2}$

So, $\frac{T_1}{T_2}=\frac{g_1}{g_1}=\left(\frac{M_2}{M_1}\right)\left(\frac{R_1}{R_2}\right)^2=\left(\frac{9}{2}\right)\left(\frac{1}{2}\right)^2=\frac{1}{8}$

6. B

Sol. Charges on the sphere after removal of sphere



$$F=\sqrt{3}\frac{Kq^2}{\ell^2}---(1)$$

$$F'=\frac{Kq^2}{\ell^2}\sqrt{\left(\frac{3}{8}\right)^2+\left(\frac{3}{4}\right)^2+\left(\frac{3}{8}\right)\left(\frac{3}{4}\right)}$$

$$F'=\frac{3\sqrt{7}}{8}F$$

7. C

Sol. There will be (N-1) minima's between two maximas if there were N slits equally spaced.

8. A

Sol. $\frac{10}{\mathrm{KA}}=2\times\frac{1}{\mathrm{A}}+\frac{8}{0.01\mathrm{A}}\Rightarrow K=0.0125$

9. D

Sol. $$\frac{\lambda}{4}=0.6r+\ell_1---(1)$$

$$(n+1)\frac{\lambda}{4}=0.6r+\ell_2----(2)$$

$$(2n+1)=\frac{0.6r+\ell_2}{0.6r+\ell_1}$$

$$(2n+1)=\frac{\frac{1}{2}+\ell_2}{\frac{1}{2}+\frac{21}{2}}$$

$$\Rightarrow \ell_2=(2n+1)11-\frac{1}{2}$$

$$\ell_2<100cm\Rightarrow n=4$$

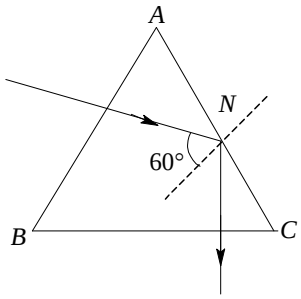
$$\ell_2=98.5\,cm$$

10. C

Sol. As the initiall potential energy will convert into rotational and translational kinetic energy so it cannot reach point C.

11. B

Sol.



At point N ray will go through total internal reflection.

12. D

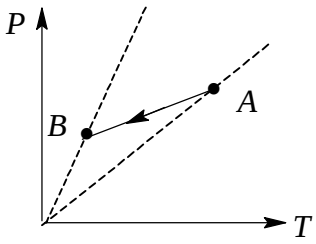
Sol. $P_1 + h\rho g = P_2 --(1)$

$P_1 A l_1 = P_0 A l_0 ---(2) \& P_2 A l_2 = P_0 A l_0 ---(3)$

$h = 25.5\times10^3 N/m^2$

13. A

Sol.



$as\ V \propto \frac{T}{P}$

$so\ V_B < V_A$

14. D

Sol. $\vec{V}_{IM} = \vec{V}_I - \vec{V}_M \Rightarrow \vec{V}_I = 10\hat{i} + \hat{j} - 2\hat{k}$

15. D

Sol. Use $\frac{\Delta g}{g} = \frac{\ell}{\Delta \ell} + \frac{2\Delta T}{T}$

16. B

Sol. $\frac{8-q}{2} + 8 - q - \frac{q}{3} = 0 \Rightarrow q = 72/11$

$\frac{V_1}{V_3} = \frac{16/11}{72/33} = \frac{2}{3}$

17. B

Sol. Nett force on block 2m is zero. So smaller block will only move.

18. A

Sol. $B=\frac{4\mu_0 I}{4\pi R}\left[\sin 45°+\sin 45°\right]=\frac{1}{2}\times 10^{-5}=5\times 10^{-6} Wb/m^2$

19. A
20. B

21. C

Sol. $m=\frac{f_0}{f_E}$ and length $\ell = f_0 + f_E$

22. B

Sol. Least count = $\frac{pitch}{no.of\ division\ on\ circular\ scale}=\frac{0.3mm}{100}=0.003mm$

Measured value $=1mm+30\times 0.003=1.09\,mm$

23. B

Sol. $E=\phi+eV_s--(1)$

$2E=\phi+eV_s^1---(2)$

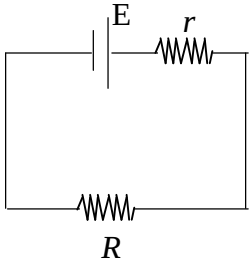
$V_s^1=9Volts$

24. C

Sol. Use Faraday Law

25. A

Sol. Equivalent circuit diagram



$where\ E=6V$ $\quad r=1\Omega$ $\quad E=\frac{E_1/r_1+E_2/r_2}{\frac{1}{r_1}+\frac{1}{r_2}}$

For maximum brightness of bulb $R=r \Rightarrow R=1\Omega$

26. A

Sol. Use conservation of momentum and Newton's experimental law.

27. B

Sol. $P_A+h\rho g-h\rho a=P_B \Leftrightarrow P_B=P_A=P_0$

28. C

Sol. $d=\sqrt{2rh}=3.2\times 10^4 meter=32km$

Area covered $=\pi d^2=3215\,km^2$

29. B

Sol. $A_r=\frac{U_0}{V_1}=\beta\frac{h_C}{R_B}\Rightarrow V_0=\beta\frac{R_C V_1}{R_B}=\frac{100\times 10\times 10^3\times 0.01}{1\times 10^3}$

$\Rightarrow V_0=10\,\text{Volt}$

30. D

Chemistry

PART – II

SECTION – A

31. A
Sol. Mg is more reactive metal.

32. D
Sol. Chains are linked by covalent bonds.

33. C
Sol. No $\beta -$ linkage in amylose and amylopectin, amylose does not have galactose also.

34. A
Sol. $HCOOH \longrightarrow H_2O(l) + CO(g)$

$2.3\,gm\left(\frac{1}{20}mole\right)$

$H_2C_2O_4 \longrightarrow H_2O(l) + CO(g) + CO_2(g)$

$\frac{1}{20}mole$

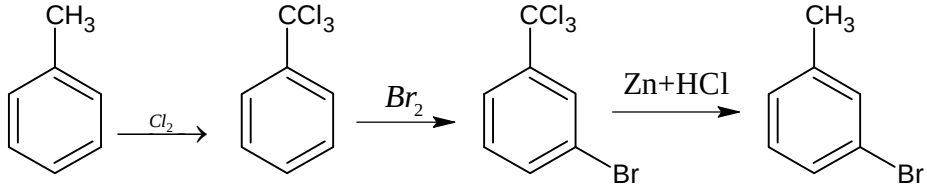
$\left(\frac{1}{20}+\frac{1}{20}\right)mole\,CO, \quad \frac{1}{20}mole\,CO_2$

CO_2 is absorbed by KOH.

So, $\frac{1}{10}$ mole of CO is left = $2.8\,\text{gm of CO}$

35. C
Sol. Facts

36. C
Sol.



37. D
Sol. MnO_4^{-2}

$Mn(VI) \longrightarrow 3d^1$

38. A
Sol. $dU = C_V dT$

$\left(\frac{dU}{dT}\right)_p = C_V$

39. A

Sol. $$\frac{d_{25^\circ C}}{d_{900^\circ C}} = \frac{\left(\frac{Z\times M}{N_A a^3}\right)_{BCC}}{\left(\frac{Z\times M}{N_A b^3}\right)_{FCC}} = \frac{\frac{2\times M}{N_A\times\left(\frac{4r}{\sqrt{3}}\right)^3}}{\frac{4\times M}{N_A\times\left(\frac{4r}{\sqrt{2}}\right)^3}} = \frac{3}{4}\sqrt{\frac{3}{2}}$$

40. A

Sol. $\Delta U = q + w$

$\Delta U = w\left(\text{for adiabatic process}\right)$

41. C

42. A

Sol. Standard state of phosphorous is $P_{\text{white,solid}}$

$P_{\text{solid ,white}} + 5O_2 \longrightarrow P_4O_{10}$ ΔH^0 is calculated by adding 2 and 3 eqn..

43. D

Sol. LDP is branched chain structure and relatively tough but flexible

44. D

Sol. Distillation under reduced pressure to decrease the boiling point.

45. A

Sol. Pyrophosphorous acid $H_4P^{III}{}_2O_5$, *p*yrophosphoric acid $H_4P^{V}{}_2O_7$

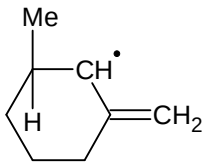
orthophosphorous acid $H_3P^{III}O_3$, Hypophosphoric acid $H_4P_2{}^{IV}O_6$

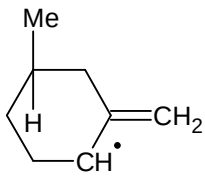
Hypophosphorous acid $H_3P^{I}O_2$

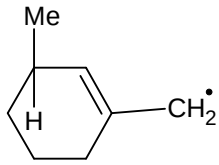
46. B

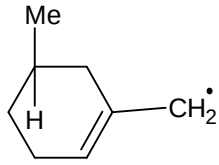
Sol. $I_2 + dil.OH^- \longrightarrow I^- + IO^- + H_2O$

47. D

Sol. 

,






48. C

Sol. $CaC_2 + 2H_2O \longrightarrow Ca\left(OH\right)_2 + C_2H_2$

$\frac{0.32}{64} = 0.005 mole = 5\, m\, mole CaC_2 \text{ or } C_2H_2$

$= 10\, m\, mole\ CO_2$

$= 224\, ml\, of\ CO_2$

49. D

Sol. $a \equiv bonding \qquad C \equiv anti\,bording$

$b \equiv bonding \qquad D \equiv non-bonding$

50. D

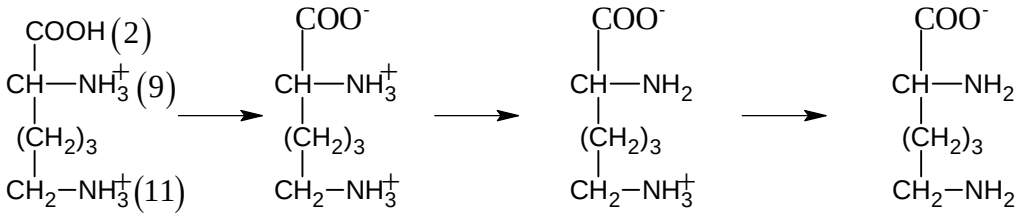
Sol. $T_c = \frac{8a}{27Rb} = 507K = 234^0C$ is more than $200°C$ .Option D is correct.

51. B

Sol. $BO_3^{3-} + CH_3OH \xrightarrow{H^+} B(OMe)_3$

52. D

Sol.

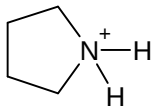


$(zwitter\,ion)$

$pK_{isoelectric} = \frac{11+9}{2} = 10$

53. D

Sol.

D is  OH^-

54. D

Sol. A is $K_2Cr_2O_7$, B is CrO_2Cl_2, C is Na_2CrO_4, D is $PbCrO_4$

55. C

Sol. Charge=0.55×60×100 C

=3300 C

=0.0342 mole of e^-

=0.0171 mole of M^{2+}

$0.017 = \frac{0.55}{MW}$, MW=32.35

56. A

Sol. $x = 6, y = 4, n = 2$

57. C

Sol. Cis - -Cis, trans - -trans, Cis- R -trans, tans –S - Cis:

58. C

Sol. $AgNO_3 \xrightarrow{\Delta} Ag + NO_2 + O_2$

$$(X) + H_2O \longrightarrow HNO_2 + HNO_3$$

$$Ag + HNO_3 \longrightarrow AgNO_3 + NO + H_2O$$

$$AgNO_3 + Na_2S_2O_3\,(excess) \longrightarrow Na_3\left[Ag\left(S_2O_3\right)_2\right] + NaNO_3$$

59. A

Sol. ii, iii, iv, v, vi, vii

60. C

Sol. $P^*_{H_2O} + P^*_{org.} = 748$

$648 + P^*_{org.} = 748, \Rightarrow P^*_{org.} = 648$

$$\chi_{org.} = \frac{P^*_{org.}}{P^*_{H_2O} + P^*_{org.}} = \frac{648}{748} = 0.1336$$

$$\frac{P^*_{H_2O} + P^*_{org.}}{P^*_{org.}} = \frac{1}{0.1336} = 7.48$$

$$1 + \frac{P^*_{H_2O}}{P^*_{org.}} = 7.48$$

$$\frac{P^*_{H_2O}}{P^*_{org.}} = 6.48$$

$$\frac{n_{H_2O}}{n_{org.}} = 6.48$$

$$\frac{1/18}{1.25/M} = 6.48$$

$$M = 146$$

Mathematics

PART – III

SECTION – A

61. C

Sol. Let $f(x) = \sin^4 x - \sin x \cos x + \cos^4 x$

$f(x) = 1 - \frac{1}{2}\sin 2x - \frac{1}{2}\sin^2 2x$

$\Rightarrow f(x) = g(t) = 1 - \frac{1}{2}t - \frac{1}{2}t^2 \quad t \in [-1, 1]$

Hence $f(x) \in \left[0, \frac{9}{8}\right]$.

62. B

Sol. $\cos\frac{\pi}{15}\cos\frac{2\pi}{15}\cos\frac{4\pi}{15}\cos\frac{7\pi}{15} = \frac{1}{16}$

$\cos\frac{3\pi}{15}\cos\frac{6\pi}{15} = \frac{1}{4}$

$\cos\frac{5\pi}{15} = \frac{1}{2}$

$\Rightarrow$ required value is $\frac{1}{128}$

63. D

Sol. $f(x) = \sqrt{3-x} - \sqrt{x+1}$ is well defined for $x \in [-1, 3]$

$f'(x) = -\frac{1}{2\sqrt{3-x}} - \frac{1}{2\sqrt{x+1}} < 0 \quad \forall\, x \in R$

$f(-1) = 2$ and $\sqrt{12-4x} = 1 + \sqrt{4x+4}$ gives $x = 1 - \frac{\sqrt{31}}{8}$

Hence $x \in \left[-1, 1 - \frac{\sqrt{31}}{8}\right)$.

64. D

Sol. Let $f(x) = \frac{\sqrt{x^2+1} + x - 1}{\sqrt{x^2+1} + x + 1}$

$\Rightarrow f(x) = \frac{x^2 + 1 - (x-1)^2}{\left(\sqrt{x^2+1} + x + 1\right)\left(\sqrt{x^2+1} - x + 1\right)}$

$f(x) = \frac{x}{\sqrt{x^2+1}+1}$

$\Rightarrow \int_1^2 \frac{x\,dx}{\sqrt{x^2+1}+1}$.

65. A

Sol. $\int \frac{e^x(x-2)}{x\left(x^2+e^x\right)} = \int \frac{3x^2+e^x+xe^x-3x^2-3e^x}{x\left(x^2+e^x\right)}dx = \int \frac{d\left(x^3+xe^x\right)}{x^3+xe^x} - 3\int \frac{dx}{x} = \ln(x^3 + xe^x) - 3\ln x + c$

$= \ln\left(\frac{x^2+e^x}{x^2}\right)+c \Rightarrow \ln\left(1+\frac{e^x}{x^2}\right)+c.$

66. C

Sol. The pair of missing numbers must be (1, 2) or (0, 3)
case 1 : (1, 2)
$\Rightarrow$ when 1 is first digit the remaining digits (1, 2, 0, 0, 5) can be arranged in 60 ways, when 5 or 2 is first digit then in 30 ways each.
case 2 : (0, 3) will have $3 \times 20 = 60$ ways
so in all we have 60 + 2.30 + 3.20 = 180 numbers.

67. C

Sol. It can be seen that K is focus of given parabola and a focal chord is divided harmonically at the focus and the point where it meets the directrix.

68. D

Sol. $\lim_{x\to\infty} \frac{x^{2013}}{e^{3x}} = 0$

$\lim_{x\to\infty}\left(\cos\frac{2}{x}\right)^{x^2} = \lim_{t\to 0} e^{-4\left(\frac{1-\cos t}{t^2}\right)} = e^{-2}$ where $t = \frac{2}{x}$

69. D

Sol. $c = a + b - 2$ and $c^2 = 2ab - 4$
$\Rightarrow a^2 + b^2 - 4a - 4b + 8 = 0$
$\Rightarrow (a - 2)^2 + (b - 2)^2 = 0$
$\Rightarrow a = b = c = 2.$

70. B

Sol. Use $\sin(A + B + C) = 0$ and $\cos^{-1}x = A$, $\cos^{-1}y = B$, $\cos^{-1}z = C$.

71. A

Sol. $I = \int_{\alpha}^{\beta} \frac{e^{f(a(\alpha-x))}}{e^{f(a(\alpha-x))}+e^{f(a(\beta-x))}}dx$ ...(1)

$I = \int_{\alpha}^{\beta} \frac{e^{f(a(x-\beta))}}{e^{f(a(\alpha-x))}+e^{f(a(\beta-x))}}dx$ ...(2)

From (1) and (2), we get

$\Rightarrow I = \frac{1}{2}(\beta-\alpha)$

72. C

Sol. The above equation reduces to $(x - 2^{2012})^2 + (x - 2^{2012}) - 2 = 0$
$\Rightarrow \alpha, \beta = 2^{2012} - 2, 2^{2012} + 1$
$\Rightarrow (\alpha - \beta)^{2014} = 3^{2014}.$

73. B

Sol. $nx(x + 1)^{n-1} = {}^nC_1x + 2{}^nC_2x^2 + \ldots\ldots + n\,{}^nC_nx^n$ ...(1)
$nx(x - 1)^{n-1} = n^nC_nx^n - (n - 1)^nC_{n-1}x^{n-1} + \ldots\ldots + (-1)^{n-1n}C_1x$ ...(2)
Again differentiating with respect to x both (1) and (2) and subtracting, we get the required result.

74. B

Sol. $\left(\frac{p+1}{2}\right)^2+\left(\frac{q+1}{2}\right)^2=\left(\frac{r+1}{2}\right)^2$

$\Rightarrow (p+1)^2+(q+1)^2=(r+1)^2$

$p+1=8,\ q+1=6,\ r+1=10$

hence minimum $(p+q+r)=21$.

75. B

Sol. Let us consider $x^3+(p-x)^3\geq p\ \forall\ x\in R$

$\Rightarrow p^3-3p^2x+3px^2\geq p$

$\Rightarrow 3p\left[\left(x-\frac{p}{2}\right)^2+\frac{p^2-4}{12}\right]\geq 0$

$\Rightarrow$ for $p<0$ $\rightarrow$ no solution

for $p=0\rightarrow$ solution exist

for $p>0\rightarrow \frac{p^2-4}{12}\geq 0$

$\Rightarrow p\in[2,\infty)\cup\{0\}$.

76. B

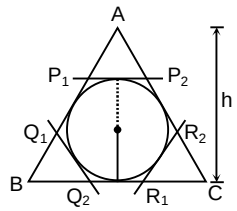
Sol. $\Delta AP_1P_2\sim\Delta ABC$

$\Rightarrow \frac{t_1}{a}=\frac{h-2r}{h}=1-\frac{2r}{h}\Rightarrow$ Hence $\frac{t_1}{a}=1-\frac{2\Delta}{sh}$

where $\Delta\equiv ar(\Delta ABC)$

$\frac{t_1}{a}=1-\frac{a}{s}\Rightarrow\frac{t_2}{b}=1-\frac{b}{s}\Rightarrow\frac{t_3}{b}=1-\frac{c}{s}$

$\Rightarrow \frac{t_1}{a}+\frac{t_2}{b}+\frac{t_3}{c}=3-\frac{(a+b+c)}{s}=3-2=1$



77. B

78. D

Sol. $d_1=\sqrt{126}$, $d_2=\sqrt{26}$

79. D

80. B

81. C

Sol. $f'(x)=e^x+3$

$f'(x)>3$

$\frac{f(x+3)-f(x-5)}{8}=f'(c)>3$

$\Rightarrow f(x+3)-f(x-5)>24$.

82. A

Sol. Put $x=y=1$

$f^2(1)+f^2(3)=2f(1)$

$\Rightarrow (f(1)-1)^2=0\Rightarrow f(1)=1$

$f(x)\ f(1)+f\left(\frac{3}{x}\right)\ f(3)=2f(x)$

$\Rightarrow f(x) = f\left(\frac{3}{x}\right) \quad \forall\, x > 0$ ...(1)

$\Rightarrow f(x)f\left(\frac{3}{x}\right) + f\left(\frac{3}{x}\right)f(x) = 2f(3)$

$\Rightarrow f(x)f\left(\frac{3}{x}\right) = 1$ ...(2)

$\Rightarrow$ from (1) and (2)

$f^2(x) = 1 \quad \forall\, x > 0$

Put $x = y = \sqrt{t}$

$f^2\left(\sqrt{t}\right) + f^2\left(\frac{3}{\sqrt{t}}\right) = 2f(t)$ or $f(t) > 0$

$\Rightarrow f(x) = 1 \quad \forall\, x > 0.$

83. A

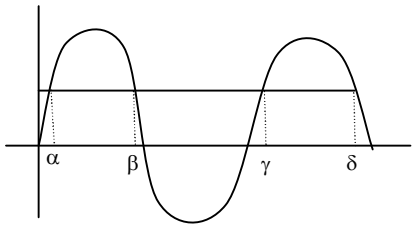
Sol. $\beta = \pi - \alpha,\ \gamma = 2\pi + \alpha,\ \delta = 3\pi - \alpha$

given expression is equal to

$$\sin\left(\frac{3\pi}{2} - \frac{\alpha}{2}\right) + 2\sin\left(\pi + \frac{\alpha}{2}\right) + 3\sin\left(\frac{\pi}{2} - \frac{\alpha}{2}\right) + 4\sin\frac{\alpha}{2}$$

$$= -\cos\frac{\alpha}{2} - 2\sin\frac{\alpha}{2} + 3\cos\frac{\alpha}{2} + 4\sin\frac{\alpha}{2}$$

$$= 2\left[\sin\frac{\alpha}{2} + \cos\frac{\alpha}{2}\right) = 2\sqrt{1+\sin\alpha} = \sqrt{5}$$



84. A

Sol. We must have

$e_1 < 1 < e_2 \Rightarrow f(1) < 0 \Rightarrow a > 3$

85. A

Sol. $1 \le \log_2(x^2 + 3x - 2) \le 3 \Rightarrow 2 \le (x^2 + 3x - 2) \le 8$

On solving them, we get $-5 \le x \le -4$ and $1 \le x \le 2$

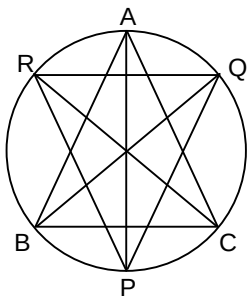
86. B

Sol. $AP = 2\sin\left(B + \frac{A}{2}\right) \Rightarrow BQ = 2\sin\left(C + \frac{B}{2}\right)$

$CR = 2\sin\left(A + \frac{C}{2}\right)$

$$\Rightarrow AP\cos\frac{A}{2} + BQ\frac{B}{2} + CR\cos\frac{C}{2} = 2(\sin A + \sin B + \sin C)$$

Maximum value of $AP\cos\frac{A}{2} + BQ\frac{B}{2} + CR\cos\frac{C}{2} = 3\sqrt{3}$

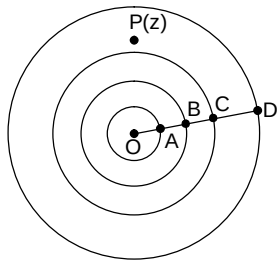


87. B

Clearly $m > \frac{1}{4}$ and $l = 0$

88. C

Sol. The minimum value of above expression will exist for $z = 5e^{i\alpha}$.
For minimum of OP + PA + PB + PC + PD
P must coincide with B.
Hence the value of the given expression is 38 units



89. C

Sol. R is not reflexive as $(2, 2) \notin R$, $(3, 3) \notin R$
Also, R is not transitive as $(2, 1) \in R$
and $(1, 2) \in R$ but $(2, 2) \notin R$.
However R is symmetric.

90. B

Sol. We have $\tan\alpha = \frac{5}{x}$ and $\tan2\alpha = \frac{30}{x}$

$\Rightarrow \tan2\alpha = \frac{30}{5\cot\alpha}$

$\Rightarrow \tan2\alpha = 6\tan\alpha \Rightarrow 3 - 3\tan^2\alpha = 1$

$\Rightarrow \tan\alpha = \sqrt{\frac{2}{3}} \Rightarrow x = 5\cot\alpha = 5\sqrt{\frac{3}{2}}$

